DESIGNING HUMAN-MACHINE INTERFACES

A focused subtopic plan for Information System Engineering

Central question: How do we translate user goals and system requirements into clear, efficient and testable interactions?

1. Recommended scope

Treat this topic as the engineering of an interface for an information system, not as a general catalogue of devices or interface trends. Explain the process of understanding users and tasks, structuring information, designing interactions, and validating the result with prototypes and usability tests.

BEST OVERALL ANGLE	**From user needs to a tested interface**

2. Keep a clear boundary with the other teams

The source PDF lists neighboring topic titles and draft subtopics. Use the boundaries below to avoid repeating their planned content.

OTHER TOPIC	THEIR PLANNED TERRITORY IN THE PDF	YOUR BOUNDARY
Input/Output devices	History and evolution of input devices and visual outputs; haptic/audio feedback; alternative and niche devices.	Do not explain device history or hardware. Discuss only the interaction requirements an interface must support.
Future of human/machine interfaces	BCI, VR/AR and spatial computing, healthcare assistive technologies, ambient computing and voice UI.	Keep the focus on established information systems and conventional interfaces. Avoid a survey of emerging technologies.
Web accessibility	WCAG and standards, inclusive design, economics of accessibility, screen readers, magnifiers and keyboard navigation.	Acknowledge accessibility as a quality requirement, but leave standards, assistive tools and detailed compliance to that group.
Design failures	Historic failures, dark patterns, deceptive design, attention economy and misleading mobile UX/touch targets.	Teach sound design practices rather than analyzing famous failures or manipulative patterns. A brief example is enough.

3. Subtopics to develop, ranked by importance

Build the presentation around the interface design process. The first five items should form the backbone; the others add useful depth without taking over the topic.

PRIORITY	SUBTOPIC	WHAT TO COVER
CORE	**1. User, task and system requirements**	Identify users and stakeholders; context of use; goals and tasks; use cases or user stories; functional requirements and usability requirements.
CORE	**2. Information architecture and navigation**	Organize content and data into clear categories and hierarchies; labels, menus, search, navigation paths and findability.
CORE	**3. Interaction and workflow design**	Translate tasks into user flows; define actions, system states, feedback, validation, error prevention, recovery and consistency.
CORE	**4. Usability and UX principles**	Visibility of system status, consistency, clear feedback, user control, recognition rather than recall, simplicity and understandable affordances.
CORE	**5. Prototyping and usability evaluation**	Sketches, wireframes and clickable prototypes; task-based user tests; observe problems; iterate and retest.
SUPPORT	**6. Cognitive ergonomics and UX laws**	Mental models, cognitive load, Hick's Law and Fitts's Law; explain how they affect choice complexity and target placement.
SUPPORT	**7. Interfaces for information-intensive tasks**	Forms, tables, filters, dashboards, data-entry validation, status indicators and clear presentation of complex information.
SUPPORT	**8. Responsive interface design**	How layouts and content adapt across screen sizes; breakpoints, content prioritization and continuity of tasks.

PRIORITY	SUBTOPIC	WHAT TO COVER
OPTIONAL	**9. Design systems and developer handoff**	Reusable components, interface patterns, design documentation, consistency and communication between designers and developers.

4. How to develop the most important subtopics

These are the sections that should receive the most explanation and examples. Connect every concept to a design decision in a real information system.

1. User, task and system requirements

Purpose: Establish what the interface must help people accomplish before deciding what it should look like.

Include: Target users and stakeholders; user goals; context of use; task analysis; use cases/user stories; functional requirements; usability goals such as efficiency, learnability and error reduction.

Useful output: A short list of user tasks and measurable acceptance criteria.

Example: For course registration, a student must find a course, check prerequisites, understand available places and complete enrolment.

2. Information architecture and navigation

Purpose: Make information easy to locate and understand.

Include: Grouping and categorization; hierarchy; meaningful labels; navigation menus; search and filters; clear paths between overview and detail pages.

Useful questions: Can users predict where information is? Are categories understandable? Can users recover when they take the wrong path?

Example: Organize a course catalogue by semester, subject area and course, with a clear route from search results to course details.

3. Interaction and workflow design

Purpose: Define how users and the system act together from start to finish.

Include: User flows; sequence of actions; system states; immediate feedback; input validation; prevention and recovery from errors; confirmation or undo where appropriate; consistent interaction patterns.

Useful output: A simple flowchart showing the main task and decision points.

Example: Search course -> open details -> check prerequisites -> select section -> confirm enrolment -> see registration status.

4. Usability and UX principles

Purpose: Turn general usability ideas into concrete design choices.

Include: Visibility of system status; consistency; clear feedback; user control; recognition rather than recall; simple language; understandable controls and affordances.

Example: Keep it applied: explain one principle, show the interface decision it suggests, and state the benefit to the user. Avoid a long list of abstract definitions.

5. Prototyping and usability evaluation

Purpose: Test whether the design works before and after implementation.

Include: Low-fidelity sketches and wireframes; clickable prototypes; representative test users; realistic tasks; observation; iteration after findings.

Useful output: Possible measures: Task completion rate, time on task, number of errors, help required and user-reported ease/satisfaction.

Example: Key message: Interface design is iterative - test, learn, revise and test again.

Cognitive ergonomics: a compact supporting section

Use mental models to explain why users expect a system to behave in a certain way; cognitive load to explain why unnecessary complexity is costly; Hick's Law to discuss choice complexity; and Fitts's Law to discuss the time needed to reach a target. Keep these as practical design explanations. Do not let them displace requirements, workflows or evaluation.

5. Lower-priority content: what not to spend much time on

Keep these areas brief. They are either outside the central engineering focus or are assigned primarily to other teams.

DE-EMPHASIZE	WHY / WHAT TO DO INSTEAD
Long history of GUI design or devices	It is not the center of information system engineering and risks overlapping with the Input/Output Devices team.
Deep technical descriptions of screens, sensors, haptics or audio	These belong mainly to the Input/Output Devices topic; refer to the device only when it affects an interaction requirement.
Detailed BCI, VR/AR, voice UI or speculative interfaces	These are explicitly assigned to Future of Human/Machine Interfaces.
Full WCAG checklist or assistive-technology tutorial	These are the focus of Web Accessibility. Mention accessibility as a requirement and cross-reference the other topic.
A catalogue of famous interface failures, dark patterns or attention-economy issues	These are assigned to Design Failures. Focus instead on the positive design process and evaluation.
Too many UX laws or abstract psychological theories	A few laws are enough when connected to concrete design decisions. Avoid theory without application.